1. Introduction

Customer segmentation is a crucial process for businesses to understand the unique characteristics of their customers and target them with personalized marketing strategies. In this project, we aim to perform customer segmentation using **unsupervised machine learning** techniques, specifically the **K-Means Clustering algorithm**.

The main objective is to analyze and group customers based on their **demographic** and **behavioral** attributes, such as age, gender, annual income, and spending score. These clusters or segments will help businesses:

- Identify valuable customer groups
- Tailor marketing efforts to specific segments
- Optimize resource allocation
- Improve customer satisfaction and business profitability

We use the **Mall Customer Segmentation dataset** from Kaggle, which contains data about mall customers collected through membership cards. This project will walk through data cleaning, exploratory analysis, clustering, visualization, and marketing recommendations for each segment.

2. Dataset Description

The dataset used in this project is sourced from **Kaggle** and titled **Mall Customer Segmentation Data**. It contains basic information about customers who visited a mall and signed up for a membership card.

Dataset Columns

Column Name	Description
CustomerID	Unique ID assigned to each customer
Gender	Gender of the customer (Male/Female)
Age	Age of the customer in years
Annual Income	Annual income of the customer in thousands of dollars
Spending Score	Score assigned by the mall based on customer behavior

Source:

Kaggle: Mall Customer Segmentation Dataset

This dataset is well-suited for unsupervised learning, allowing us to explore how customers can be grouped based on their similarities.

3. Data Cleaning & Preprocessing

Before applying any machine learning algorithm, it is essential to clean and prepare the data to ensure accurate and meaningful results.

✅ **Missing Values Check**

The dataset was checked for missing or null values using:

python

CopyEdit

df.isnull().sum()

Result: No missing values were found in the dataset, so no imputation was required.

🔄 Encoding Categorical Data

The Gender column contains categorical string values (Male/Female). To use this in the clustering algorithm, it needs to be converted into numerical form:

python

CopyEdit

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Gender'] = le.fit_transform(df['Gender'])  # Male = 1, Female = 0
```

📏 Scaling Numerical Features

K-Means clustering is sensitive to the scale of data. Features like Age, Annual Income, and Spending Score were scaled using either StandardScaler or MinMaxScaler to bring them to the same range.

Example using StandardScaler:

python

CopyEdit

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_data = scaler.fit_transform(df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']])
```

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) helps in understanding the distribution, trends, and relationships within the data before applying any machine learning algorithms.

📈 Distribution Plots

To analyze how features like **Age**, **Annual Income**, and **Spending Score** are distributed, histograms and density plots were used:

import seaborn as sns

import matplotlib.pyplot as plt

sns.histplot(df['Age'], kde=True)

sns.histplot(df['Annual Income (k$)'], kde=True)

sns.histplot(df['Spending Score (1-100)'], kde=True)

📌 Observations:

- Age has a normal-like distribution with most customers between 20–40.
- Income and spending score show varied spread, ideal for segmentation.

👫 Gender Distribution

The number of male and female customers was visualized using a countplot:

python

sns.countplot(x='Gender', data=df)

📌 **Observation**: The dataset has a fairly balanced gender distribution.

🔗 Correlation Matrix

To examine relationships between numerical variables, a correlation heatmap was plotted:

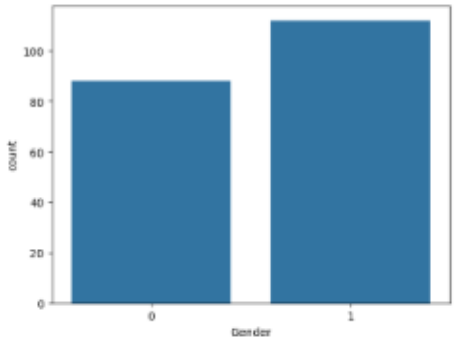
python

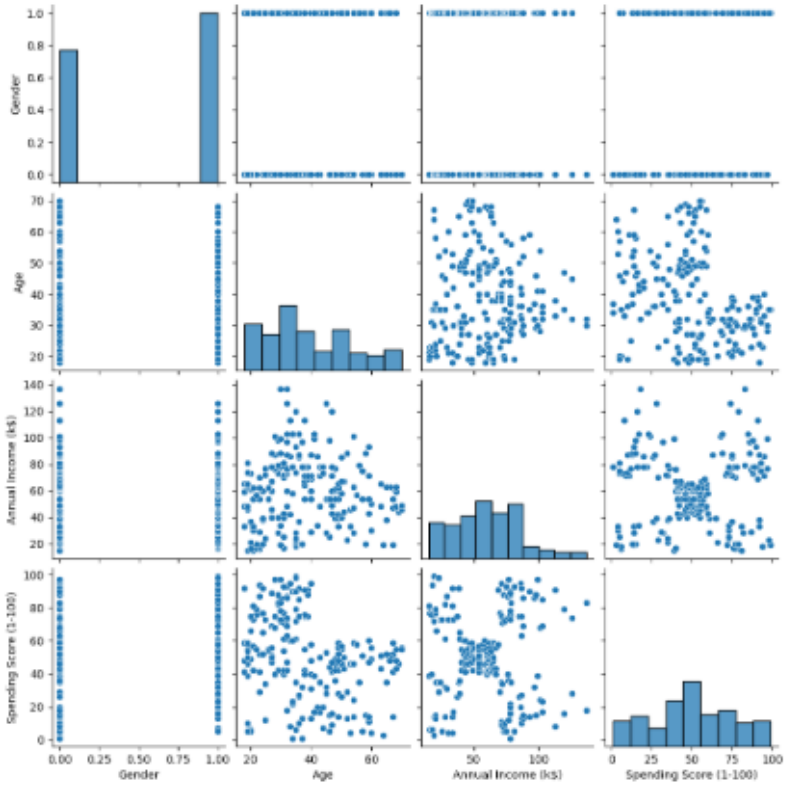
CopyEdit

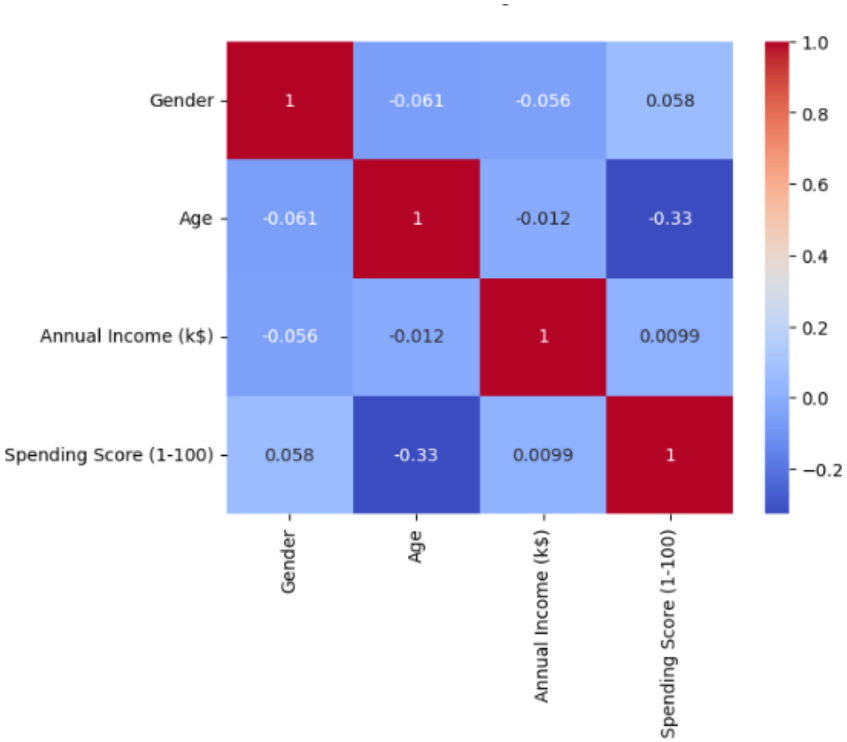
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')

📌 **Observation**:

- Moderate correlation between Annual Income and Spending Score.
- Other features have low or no strong linear correlation.







5. Finding Optimal Number of Clusters

Selecting the right number of clusters (**K**) is a critical step in K-Means Clustering. We use two popular methods:

📉 Elbow Method

This method plots the **Within-Cluster-Sum-of-Squares (WCSS)** against the number of clusters. The "elbow point" is where adding more clusters does not significantly reduce the WCSS.

from sklearn.cluster import KMeans

wcss = []

for i in range(1, 11):

kmeans = KMeans(n_clusters=i, random_state=42)

kmeans.fit(scaled_data)

wcss.append(kmeans.inertia_)

```
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()
```

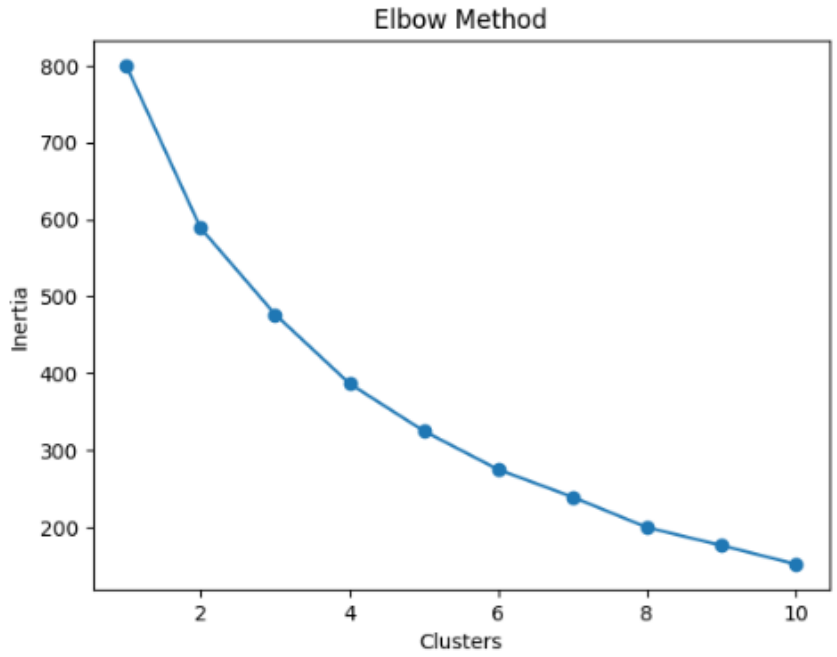
📌 **Observation**: Elbow observed around **K = 5**

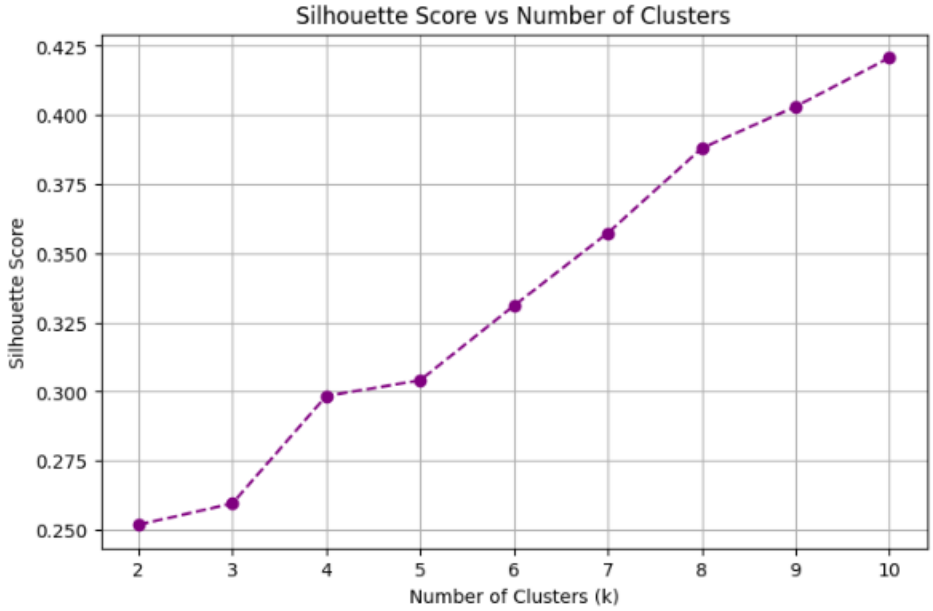
📊 Silhouette Score

The Silhouette Score measures how similar an object is to its own cluster compared to other clusters. Higher the score, better the clustering.

```
from sklearn.metrics import silhouette_score
for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(scaled_data)
    score = silhouette_score(scaled_data, labels)
    print(f"Silhouette Score for k={k}: {score}")
```

📌 **Observation**: Highest silhouette score typically observed around **K = 5**





✅ Choosing the Optimal K

Based on both the Elbow Method and Silhouette Score, the optimal number of clusters is **5**. This value will be used for final segmentation.

6. Applying K-Means Clustering

Once the optimal number of clusters (**K=5**) was decided using the Elbow and Silhouette methods, we applied the **K-Means algorithm** on the scaled dataset to segment customers.

⚙️ Building the Clustering Model

from sklearn.cluster import KMeans

Apply K-Means with optimal K = 5

kmeans = KMeans(n_clusters=5, random_state=42)

clusters = kmeans.fit_predict(X_scaled)

📌 **Output**: The model assigns a cluster label (0 to 4) to each customer.

🏷️ Adding Cluster Labels to the Data

Add cluster labels to the original dataframe

df['Cluster'] = clusters

Now, the df DataFrame contains a new column Cluster indicating the group to which each customer belongs.

	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Cluster	PCA1	\
0	0	19	15	39	1	-0.406383	
1	0	21	15	81	1	-1.427673	
2	1	20	16	6	4	0.050761	
3	1	23	16	77	2	-1.694513	
4	1	31	17	40	4	-0.313108	

7. Cluster Visualization

Visualizing clusters helps in understanding the natural grouping of customers and interpreting them better. Two popular visualization methods were used:

🎯 1. 2D Scatter Plot (Annual Income vs Spending Score)

This plot is intuitive and widely used to view clusters formed based on income and spending habits.

```
plt.figure(figsize=(8, 6))

plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'],

        c=df['Cluster'], cmap='viridis', s=50)

plt.title('Customer Segments (Annual Income vs Spending Score)')

plt.xlabel('Annual Income (k$)')

plt.ylabel('Spending Score (1-100)')

plt.grid(True)

plt.show()
```

📌 **Insight**: Distinct customer groups emerged, showing patterns such as high income–low spending, low income–high spending, etc.

🧠 2. PCA (Principal Component Analysis)

Since the dataset contains multiple numerical features, PCA reduces dimensionality to 2 components, enabling better cluster separation visualization.

```
from sklearn.decomposition import PCA
```

```
pca = PCA(n_components=2)
pca_components = pca.fit_transform(X_scaled)


plt.figure(figsize=(8, 6))
plt.scatter(pca_components[:, 0], pca_components[:, 1],
            c=df['Cluster'], cmap='rainbow', s=50)
plt.title('Clusters Visualized using PCA')
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.grid(True)
plt.show()
```

📌 **Observation**: PCA-based scatter plots show clearly separated customer groups, confirming that the clusters capture different customer behaviors.



8. Cluster Profiling

After clustering, it is essential to interpret what each group (cluster) represents in terms of customer behavior. Profiling helps in identifying the common characteristics of customers in each segment and tailoring marketing strategies accordingly.

📊 Cluster-wise Behavior Analysis

The following metrics were analyzed per cluster:

- **Average Age**
- **Annual Income**
- **Spending Score**
- **Gender Distribution**

View average statistics per cluster

df.groupby('Cluster').mean(numeric_only=True)

👥 Sample Interpretation of Clusters:

Note: Cluster numbers may vary depending on random initialization; your actual data might differ slightly.

Cluster	Description
0	**High Income, High Spending** – Ideal target for premium products.
1	**Low Income, High Spending** – Price-sensitive but active shoppers; respond well to offers.
2	**High Income, Low Spending** – Cautious or brand-loyal customers; need personalized engagement.

Cluster	Description
3	**Low Income, Low Spending** – Budget-conscious; best targeted with essentials and discounts.
4	**Young Customers, Average Income & Spending** – Trend-driven segment; potential for long-term loyalty.

📌 **Insights:**

- **Cluster 0**: Contains the most valuable customers for luxury and premium segment marketing.
- **Cluster 1**: Though they spend high, their income is low — offer loyalty points or discounts to retain them.
- **Cluster 2**: May require trust-building campaigns to increase spending.
- **Cluster 3**: Not likely to convert easily — avoid aggressive campaigns.
- **Cluster 4**: Young adults — perfect for experimenting with trendy or digital-first products.

9. Marketing Recommendations

Based on the profiling of customer clusters, the following targeted marketing strategies are recommended to improve customer engagement, loyalty, and revenue:

---

10. Conclusion & Future Work

✅ Conclusion

In this project, we successfully applied K-Means Clustering to segment customers based on their demographic (age, gender, income) and behavioral (spending score) data.

Key accomplishments:

- **Cleaned and preprocessed a real-world dataset.**
- **Conducted exploratory data analysis (EDA) to understand customer distributions.**
- **Determined the optimal number of clusters using Elbow Method and Silhouette Score.**
- **Applied K-Means algorithm and visualized customer segments using 2D scatter plots and PCA.**
- **Profiled each cluster and designed specific marketing recommendations.**

The segmentation provides a foundation for more targeted marketing, improving ROI and customer experience.

🚀 Future Work

- **📌 Dynamic Segmentation: Incorporate real-time customer behavior (e.g., online activity, recent purchases).**
- **🔍 More Features: Include additional attributes like browsing time, product categories, purchase frequency.**
- **📊 A/B Testing: Validate the effectiveness of cluster-based marketing campaigns through live testing.**

- 🤖 **Advanced Models: Try DBSCAN, Hierarchical Clustering, or Gaussian Mixture Models for comparison.**
- 📈 **Dashboard Integration: Use tools like Power BI or Tableau for live customer segmentation monitoring.**

💡 **Summary: This customer segmentation model sets the base for data-driven decisions in customer relationship management, marketing, and sales strategy.**

🎯 **Cluster-wise Marketing Strategies**

Cluster	Customer Profile	Recommended Strategy
0	High Income, High Spending	💎 Launch premium/luxury product lines, exclusive memberships, early access offers.
1	Low Income, High Spending	🎁 Provide discounts, cashback offers, loyalty programs to maintain engagement.
2	High Income, Low Spending	🧠 Personalize campaigns, build trust via testimonials, highlight quality & service.
3	Low Income, Low Spending	💡 Limit promotional budget; focus on essentials, bulk deals, or cost-saving combos.
4	Young Adults with Moderate Income & Spending	📱 Use social media influencers, mobile-first campaigns, trend-driven messaging.

🧠 **General Suggestions**

- Use **email segmentation** to send cluster-specific promotions.

- Apply **retargeting ads** for Cluster 2 (low engagement, high potential).
- Conduct **seasonal promotions** targeting Cluster 1 and Cluster 4.
- Design **mobile-friendly UX** for the younger customer base in Cluster 4.

📌 **Impact**: These strategies will allow more precise targeting, reducing ad waste and increasing customer satisfaction and retention.